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Technical Presentation

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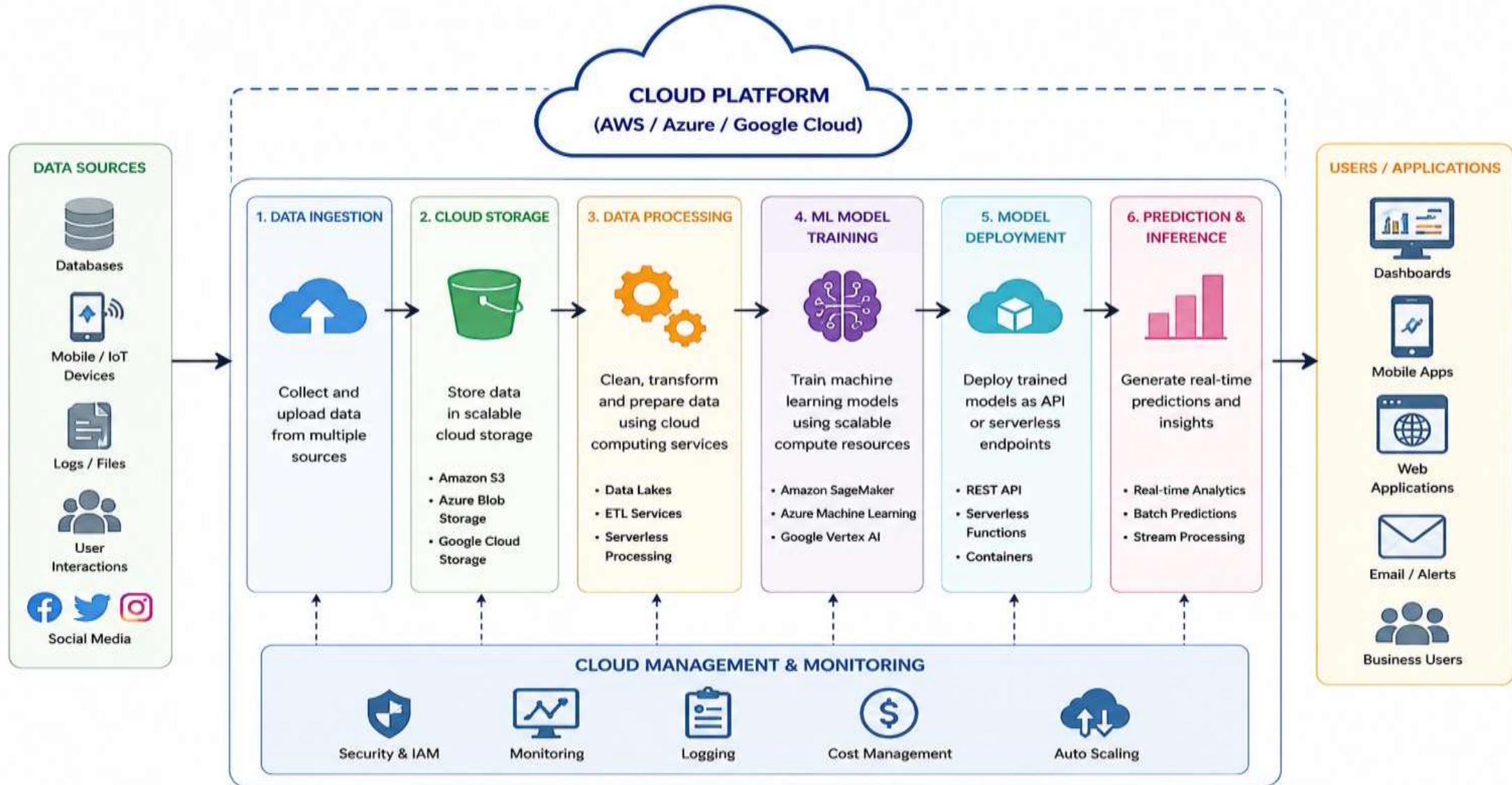
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Cloud Computing

Introduction

Artificial Intelligence (AI) and Machine Learning (ML) are transforming the way organizations use cloud computing. By combining intelligent algorithms with the cloud's scalable computing power, businesses can analyze massive amounts of data, automate tasks, and deliver smarter services across industries.

This presentation explains how AI and ML integrate with cloud platforms, their working process, applications, benefits, and real-world use cases.

AI & ML in Cloud Platform – Architecture



How It Works

1. Collect data from users or devices.
2. Store data in cloud storage.
3. Process the data.
4. Train ML models using cloud resources.
5. Deploy the model.
6. Generate predictions.
7. Display results to users.

Industry Relevance

AI and ML in cloud platforms are widely used in:

Healthcare

Banking & Finance

Education

E-commerce

Transportation

Benefits

Scalability

Lower infrastructure cost

High computing power

Real-time analytics

Real-World Example

Netflix Recommendation System

Netflix uses Artificial Intelligence (AI) and Machine Learning (ML) on cloud platforms to recommend movies and TV shows based on users' viewing history, search behavior, ratings, and preferences. The cloud stores and processes large amounts of data, while ML algorithms analyze user behavior to provide personalized recommendations. This improves user experience, increases customer satisfaction, and helps Netflix retain more subscribers.

Key Points

- ☐ Collects user viewing history and preferences.
- ☐ Stores data on cloud servers.
- ☐ ML analyzes user behavior.
- ☐ AI recommends personalized movies and TV shows.
- ☐ Improves customer satisfaction and watch time.

Other examples:

- ☐ Amazon – Product recommendations
- ☐ Google Photos – Image recognition
- ☐ Microsoft Azure AI – Chatbots and speech recognition

Advantages & Challenges

Advantages

- ❑ AI and ML automate repetitive tasks and improve efficiency.
- ❑ Cloud platforms provide scalable computing resources for AI applications.
- ❑ AI enables faster and more accurate decision-making.

Challenges

- ❑ Protecting sensitive data and ensuring privacy is a major concern.
- ❑ Training AI models requires large amounts of high-quality data.
- ❑ Running advanced AI models in the cloud can be expensive.

Conclusion

- ❑ AI and ML become more powerful when combined with cloud computing.
- ❑ Cloud platforms provide storage, computing power, and AI tools.
- ❑ Organizations use cloud AI to improve efficiency, automate tasks, and make smarter decisions.

Key Takeaway

- ❑ Cloud computing enables AI and ML to be accessible, scalable, and affordable for businesses of all sizes.

References

- ❑ Google Cloud AI
- ❑ Microsoft Azure AI
- ❑ Amazon Web Services (AWS) AI
- ❑ IBM Cloud AI
- ❑ Research papers from IEEE Xplore

The image features a soft, pink watercolor wash as a background. The wash is composed of several overlapping, irregular strokes of varying shades of pink, creating a textured, painterly effect. The colors range from a pale, almost white pink to a deeper, more saturated magenta. The edges of the wash are soft and feathered, blending into the white background. Centered over this wash is the text "Thank You" written in a black, elegant cursive script. The word "Thank" is on the top line, and "You" is on the bottom line, with the 'Y' in "You" being particularly large and stylized, with a long, sweeping tail that curves under the word. The overall composition is simple and elegant, with a warm and appreciative feel.

Thank
You



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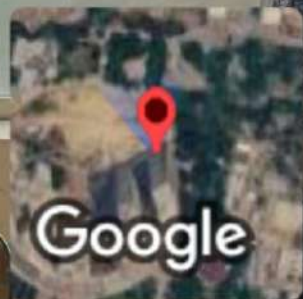
The Role of Artificial Intelligence and Machine Learning in Cloud Platforms

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Cloud Computing and Big Data Analytics



GPS Map Camera



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